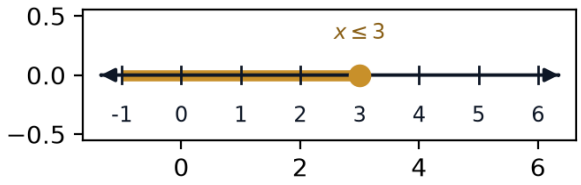
Linear Functions & Inequalities

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • [FRQ] • Target: Solving One-Variable Linear Inequalities

Solve each linear inequality for x and express the solution set in interval notation:

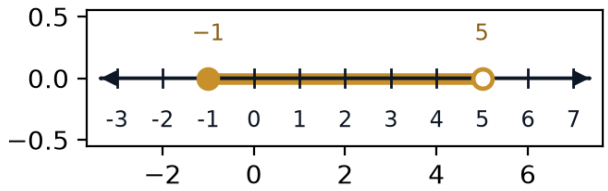


a $7 - 4(2x - 3) \geq 3x - 14$

b $-9 < \frac{5 - 3x}{2} \leq 6$

2 [Foundational] • [FRQ] • Target: Compound Inequalities & Number Line Graphs

Solve the compound inequality $-5 \leq 2x - 3 < 7$ and verify that the number line below accurately represents its solution set.



3 [Foundational] • [FRQ] • Target: Solving Systems by Substitution

Solve the linear system by substitution and state the value of $x + y$: $y = 3x - 7$ and $4x + 3y = 31$.

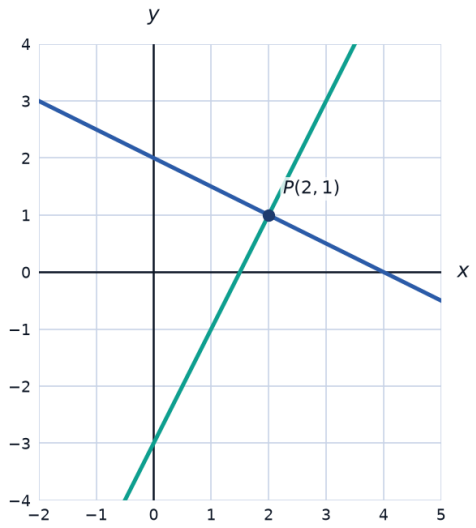
4 [Foundational] • [FRQ] • Target: Solving Systems by Elimination

Solve the linear system by elimination: $5x - 2y = 26$ and $3x + 4y = 10$.

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5 [Foundational] • [FRQ] • Target: Graphical Intersection of Linear Systems

The graphs of $y = 2x - 3$ and $y = -\frac{1}{2}x + 2$ are shown in the xy-plane. Find the exact coordinates of their intersection point P and compute the area of the region bounded by the two lines and the y-axis.



6 [Foundational] • [FRQ] • Target: Two-Variable System Word Problems

A university lecture hall has 320 seats divided between standard seats (25 dollars each) and VIP balcony seats (45 dollars each). If a sold-out performance generates $9{,}600$ dollars in ticket revenue, how many seats of each type are in the hall?

7 [Foundational] • [FRQ] • Target: Verifying Solutions to Systems of Inequalities

Determine whether the point $(3, -2)$ lies in the solution region of the system of inequalities $2x - 3y > 10$ and $x + 4y \leq -5$. Justify algebraically.

Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

8 [Intermediate] • [FRQ] • Target: Dependent Systems (Infinitely Many Solutions)

Find the values of the constants a and b such that the linear system $ax + 6y = 15$ and $8x - 12y = b$ has infinitely many solutions.

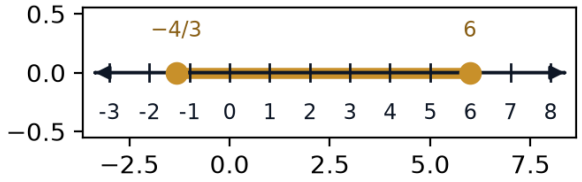
9 [Intermediate] • [FRQ] • Target: Inconsistent Systems (No Solution)

For what value of k does the linear system $(k - 2)x + 4y = 9$ and $3x + (k + 2)y = 5$ have no solution?

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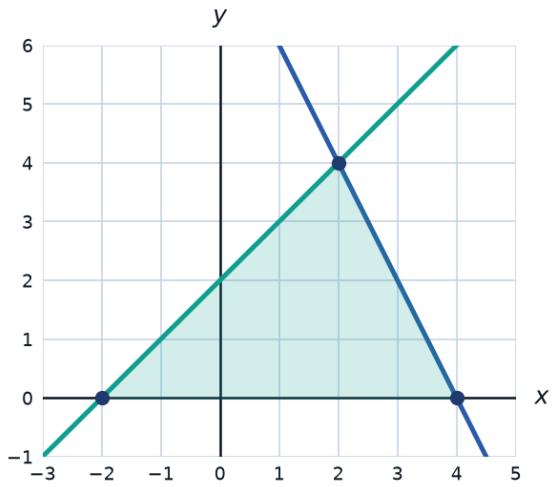
10 [Intermediate] • [FRQ] • Target: Absolute Value Inequalities & Integer Solutions

Solve the absolute value inequality $|3x - 7| + 4 \le 15$, express the solution set as an interval, and list all integer solutions.



11 [Intermediate] • [FRQ] • Target: Graphing & Area of Feasible Regions

The shaded region in the xy-plane represents the feasible set of the system $y \ge 0$, $y \le x + 2$, and $y \le -2x + 8$. Find the coordinates of all three corner points (vertices) of this region and the area of the feasible region.



12 [Intermediate] • [FRQ] • Target: Linear Programming Optimization in Context

A coffee roaster blends Ethiopian beans (14 dollars/kg) and Colombian beans (10 dollars/kg) to make a batch of at least 40 kg whose total ingredient cost is at most 480 dollars. Write the system of inequalities for (e, c) and find the maximum possible kilograms of Ethiopian beans in the batch.

13 [Intermediate] • [FRQ] • Target: Structural Linear Combination Shortcut

Without solving for x and y individually, find the value of $x - y$ given the system $7x + 3y = 41$ and $5x + 5y = 35$.

14 [Intermediate] • [FRQ] • Target: Relative Velocity Linear Systems

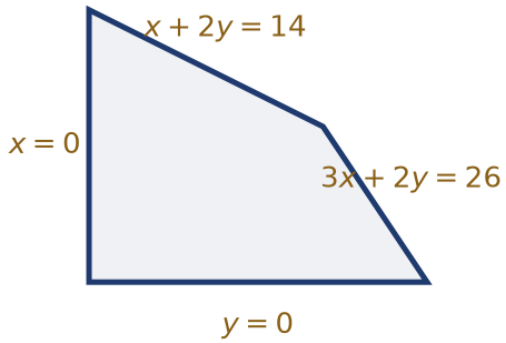
A motorboat travels 48 km downstream in 1.5 hours and returns the same 48 km upstream in 2.4 hours. Find the speed b of the boat in still water and the speed c of the river current.

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Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

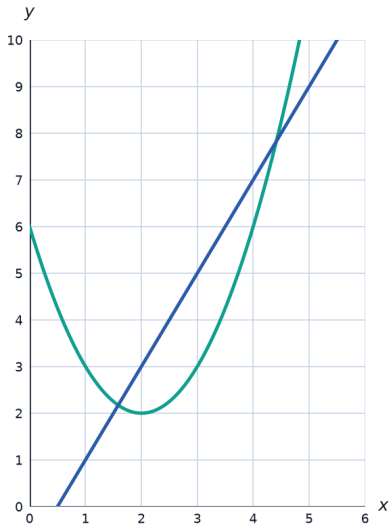
15 [Challenging] • [FRQ] • Target: Corner-Point Theorem & Linear Programming • Cross-Lesson: Topic 8.01 (Polygonal Regions & Coordinate Geometry)

Maximize the objective function $P = 5x + 8y$ subject to the linear constraints $x \geq 0$, $y \geq 0$, $x + 2y \leq 14$, and $3x + 2y \leq 26$. State the optimal point (x^*, y^*) and the maximum value $P_{\max}$.



16 [Challenging] • [FRQ] • Target: Linear-Quadratic System & Chord Length • Cross-Lesson: Topic 3.01 (Quadratics) & Topic 7.01 (Distance Formula)

Find all points (x, y) where the line $2x - y = 1$ intersects the quadratic curve $y = x^2 - 4x + 6$, and calculate the exact length of the chord connecting the intersection points.



17 [Challenging] • [FRQ] • Target: Piecewise Absolute-Value System Multiplicity • Cross-Lesson: Topic 1.01 (Absolute Value Functions)

For what values of c does the system $y = |x - 2| + 1$ and $y = c$ have (a) zero solutions, (b) exactly one solution, and (c) two distinct solutions?

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18 [Challenging] • [FRQ] • Target: Symmetric Three-Variable Linear Systems • Cross-Lesson: Topic 10.01 (Algebraic Structure & Symmetry)

Solve the three-variable linear system $x + y = 7$, $y + z = 11$, and $z + x = 10$, and compute the product xyz.

19 [Challenging] • [FRQ] • Target: Inscribed Circle of a Linear Feasibility Triangle • Cross-Lesson: Topic 9.01 (Circles & Inradius Formula)

The system of inequalities $x \geq 1$, $y \geq 2$, and $3x + 4y \leq 29$ defines a right triangle in the xy-plane. Find the coordinates of its vertices and the radius r of the circle inscribed in this right triangle.

20 [Challenging] • [FRQ] • Target: Two-Component Percentage Composition System • Cross-Lesson: Topic 5.01 (Percentages & Proportions)

A nutritionist prepares a supplement mix from Powder A (30% protein, 10% fiber) and Powder B (15% protein, 25% fiber). To get exactly $180\,\text{g}$ of protein and $110\,\text{g}$ of fiber, how many grams of Powder A and Powder B are required?

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

21 [Foundational] • [MCQ] • Target: Direct Elimination in Linear Systems

From the governing equation where if $3x + 2y = 19$ and $x - 2y = 1$, what is the value of x?

a. 3

b. 4

c. 5

d. 6

22 [Foundational] • [MCQ] • Target: Sign Reversal in Inequalities

Which of the following is the solution set to the inequality $5 - 3x > 17$?

a. $x < -4$

b. $x > -4$

c. $x < -\frac{22}{3}$

d. $x > -\frac{22}{3}$

23 [Foundational] • [MCQ] • Target: Solving Linear Systems

If (x, y) is the solution to the system $2x + 5y = 24$ and $3x + 2y = 14$, what is the value of $x + y$?

a. 5

b. 6

c. 7

d. 8

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24 [Foundational] • [MCQ] • Target: Translating Verbal Constraints into Inequalities

A bakery sells croissants for 3.50 dollars each and muffins for 2.25 dollars each. Maya buys a total of 12 pastries and spends at most 36.00 dollars. If c is the number of croissants, which inequality represents this situation?

a. $3.50c + 2.25(12 - c) \le 36.00$

b. $3.50c + 2.25(12 - c) \ge 36.00$

c. $3.50(12 - c) + 2.25c \le 36.00$

d. $3.50c + 2.25c \le 36.00$

25 [Foundational] • [MCQ] • Target: Testing Points in Inequality Systems

Which point (x, y) lies in the solution set of the system $y > 2x - 1$ and $y < -x + 5$?

a. $(0, -2)$

b. $(1, 2)$

c. $(3, 4)$

d. $(4, 2)$

26 [Foundational] • [MCQ] • Target: Collinear Systems

How many solutions (x, y) does the system $4x - 6y = 10$ and $-6x + 9y = -15$ have?

a. Zero

b. Exactly one

c. Exactly two

d. Infinitely many

27 [Foundational] • [MCQ] • Target: Greatest Integer Solutions to Inequalities

What is the greatest integer x that satisfies the inequality $\frac{4x + 3}{5} < 7$?

a. 6

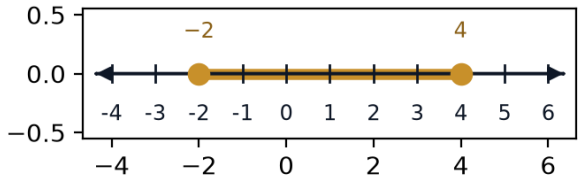
b. 7

c. 8

d. 9

28 [Foundational] • [MCQ] • Target: Matching Number Lines to Absolute Value Inequalities

The number line below shows the solution set to which inequality?



a. $|x - 1| \le 3$

b. $|x + 1| \le 3$

c. $|x - 3| \le 1$

d. $|x - 1| \ge 3$

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Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

29 **[Intermediate]** • **[MCQ]** • **Target: No-Solution Condition for Linear Systems**

In the system of equations $kx - 3y = 8$ and $12x - 9y = 20$, k is a constant. If the system has no solution, what is the value of k?

a. -4

b. 3

c. 4

d. 6

30 **[Intermediate]** • **[MCQ]** • **Target: Symmetric Linear Combination**

Given the simultaneous linear system $7x - 4y = 23$ and $3x + 8y = 5$, compute the linear combination $5x + 2y$.

a. 12

b. 14

c. 16

d. 18

31 **[Intermediate]** • **[MCQ]** • **Target: Two-Constraint Resource Allocation**

A landscaping crew can mow a small lawn in 40 minutes and a large lawn in 75 minutes. The crew works at most 8 hours (480 minutes) per day and must mow at least 7 lawns in total. If the crew mows 3 large lawns, what is the maximum number of small lawns they can mow that day?

a. 4

b. 5

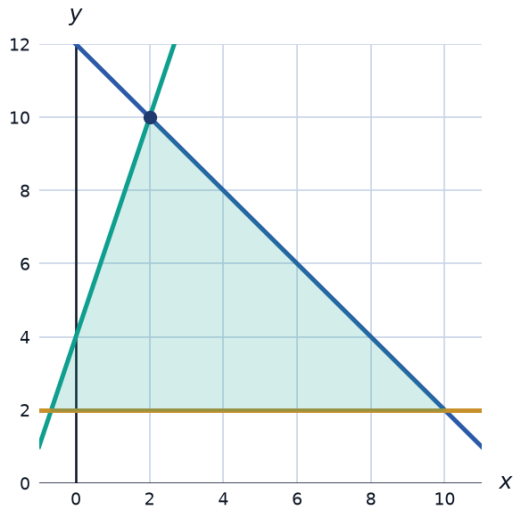
c. 6

d. 7

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32 [Intermediate] • [MCQ] • Target: Vertex Identification in Linear Inequalities

In the xy-plane, the solution set of the system $y \leq 3x + 4$, $y \leq -x + 12$, and $y \geq 2$ is a triangular region. What is the x-coordinate of the vertex with the greatest y-coordinate?



a. 1.5

b. 2

c. 2.5

d. 3

33 [Intermediate] • [MCQ] • Target: Counting Integer Solutions

For how many integer values of x is the compound inequality $11 < 4x + 3 < 35$ satisfied?

a. 4

b. 5

c. 6

d. 7

34 [Intermediate] • [MCQ] • Target: Intersecting Inequality Constraints

A chemistry lab needs between 300 mL and 450 mL of acid mixture. Reagent X contains 20% acid and Reagent Y contains 50% acid. If the lab uses 200 mL of Reagent X and y mL of Reagent Y, and the total pure acid must be at least 115 mL, what is the valid range for y?

a. $100 \leq y \leq 250$

b. $120 \leq y \leq 250$

c. $150 \leq y \leq 250$

d. $150 \leq y \leq 300$

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35 [Intermediate] • [MCQ] • Target: Rational-Form Linear Systems

Suppose two real numbers satisfy $\frac{x+y}{x-y} = 5$ and $x + y = 10$. Determine the product xy.

a. 16

b. 20

c. 24

d. 25

36 [Intermediate] • [MCQ] • Target: Quadrant Analysis of Linear Inequalities

In the xy-plane, which quadrants contain NO points (x, y) satisfying the system $y > 2x + 3$ and $y > -\frac{1}{2}x + 1$?

a. Quadrant III only

b. Quadrant IV only

c. Quadrants III and IV

d. Quadrants II and IV

37 [Intermediate] • [MCQ] • Target: System Word Problems for Unit Prices

The cost of 4 notebooks and 3 pens is 24.25 dollars, and the cost of 2 notebooks and 5 pens is 18.25 dollars. What is the cost of 1 notebook and 1 pen combined?

a. 5.75 dollars

b. 6.25 dollars

c. 6.50 dollars

d. 6.75 dollars

Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Lattice Points in a Feasible Region • Cross-Lesson: Topic 6.01 (Combinatorics & Counting)

How many ordered pairs of integers (x, y) satisfy the strict inequalities $x > 0$, $y > 0$, and $2x + 3y < 14$?

a. 7

b. 8

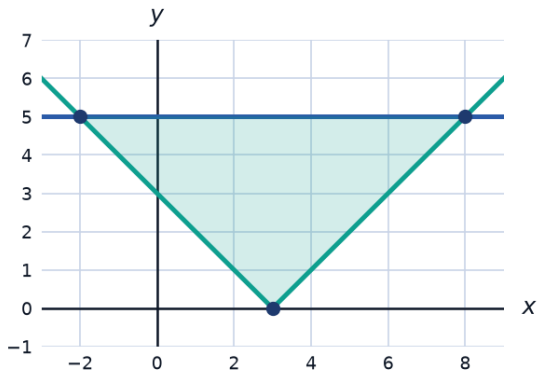
c. 9

d. 10

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39 [Challenging] • [MCQ] • Target: Area & Perimeter of Absolute-Value Regions • Cross-Lesson: Topic 7.01 (Isosceles Right Triangles & Special Ratios)

In the xy-plane, the system of inequalities $y \geq |x - 3|$ and $y \leq 5$ encloses a triangular region. What is the perimeter of this region?



a. $10 + 5\sqrt{2}$

b. $10 + 10\sqrt{2}$

c. 20

d. $20\sqrt{2}$

40 [Challenging] • [MCQ] • Target: Determinant Condition for Unique Solutions • Cross-Lesson: Topic 10.01 (Algebraic Structure)

If the linear system $x + ky = 5$ and $kx + 4y = 10$ has a unique solution (x, y), which of the following CANNOT be the value of k?

a. -2 only

b. 2 only

c. 0

d. -2 and 2

41 [Challenging] • [MCQ] • Target: Geometric Probability on a Feasible Region • Cross-Lesson: Topic 6.01 (Probability) & Topic 8.01 (Area)

A point (x, y) is chosen uniformly at random from the rectangle $0 \leq x \leq 6$, $0 \leq y \leq 4$. What is the probability that the point satisfies $x + y \leq 4$?

a. $\frac{1}{4}$

b. $\frac{1}{3}$

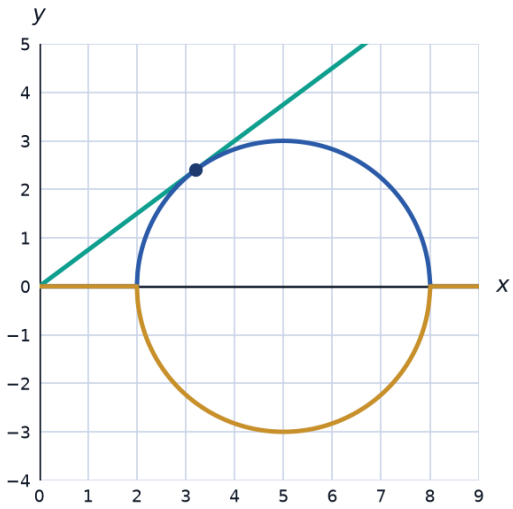
c. $\frac{1}{2}$

d. $\frac{2}{3}$

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42 [Challenging] • [MCQ] • Target: Linear-Circle Tangency System • Cross-Lesson: Topic 9.01 (Circles & Tangent Lines)

For what value of $m > 0$ does the system $y = mx$ and $x^2 + y^2 - 10x + 16 = 0$ have exactly one real solution (x, y)?



a. $\frac{3}{4}$

b. $\frac{4}{3}$

c. $\frac{3}{5}$

d. $\frac{5}{4}$

43 [Challenging] • [MCQ] • Target: Linear Systems in Transformed Variables • Cross-Lesson: Topic 4.01 (Exponentials & Radicals)

If a and b are positive real numbers satisfying $\log_2(a) + \log_2(b) = 5$ and $3\log_2(a) - \log_2(b) = 7$, what is the value of $\frac{a}{b}$?

a. 1

b. 2

c. 4

d. 8

44 [Challenging] • [MCQ] • Target: Absolute Value Diamond & Half-Plane Area • Cross-Lesson: Topic 8.01 (Area & Coordinate Symmetry)

The region defined by $|x| + |y| \le 6$ in the xy-plane is a square. What is the area of the portion of this square that also satisfies $y \ge x - 2$?

a. 36

b. 48

c. 54

d. 60

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45 [Challenging] • [MCQ] • Target: Industrial Production Linear System • Cross-Lesson: Topic 5.01 (Rates & Proportions)

A factory produces two models of solar panels, Model X and Model Y. Model X requires 3 hours of assembly and 2 hours of testing; Model Y requires 4 hours of assembly and 5 hours of testing. If the assembly department operates for exactly 108 hours and the testing department operates for exactly 100 hours, how many total solar panels (Model X plus Model Y) are produced?

a. 28

b. 30

c. 32

d. 35

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